Lecture population balances

Tuesday, 4 April 2023 12:00 PM

→ Steady state cont. extraction in well-stirred vessel

Assumptions:
- Negligible coalescence
- Dispersion phase is small : "lean"
- Feed drops of identical size ; drop breakage is binary & equal
- Conc. changes in drops by diffusion
- Cont. phase (lean no solute) has near zero conc. & changes negligibly due to MT

Solute diffuses out of the continuous phase ↑ → need surface flux (rate @ which solid diffuses out)

Assume you know the age of the drop, & the average conc. inside the droplet

size of the drop

→ Smaller droplets take longer time

→ Infinite breakage rate model → breakup occurs immediately

→ Steady state drop-size distribution { Generations of droplets { 0th → q_0 enters (w/o breakup just transferring solute) → if these escaped quickly then there is not much conc. difference/loss in these drops
1st → q_0 breaks in q_1
2nd → q_1 breaks into further

MC → generate random no. & let it play w/ time (Markov chain

Aggregation Model (Coalescence) → blending interior of particles

→ just grouping together

Aggregation frequency: $a(x,y)$ → fraction of particle pair (x,y) that aggregate per unit time

$$a(x,y) = a(y,x)$$

Only aggregation (No breakup) : $n(v,t)$; v → particle volume

+ We care about all particles that can come together to form agglomerate

↳ Population balance = Mass balance

Total w.r.t dN

Laplace transform → for when you want to deconvolute two integral expression